

Introduction to Earth System

- Temperature
- Wind
- Pressure

I. Temperature

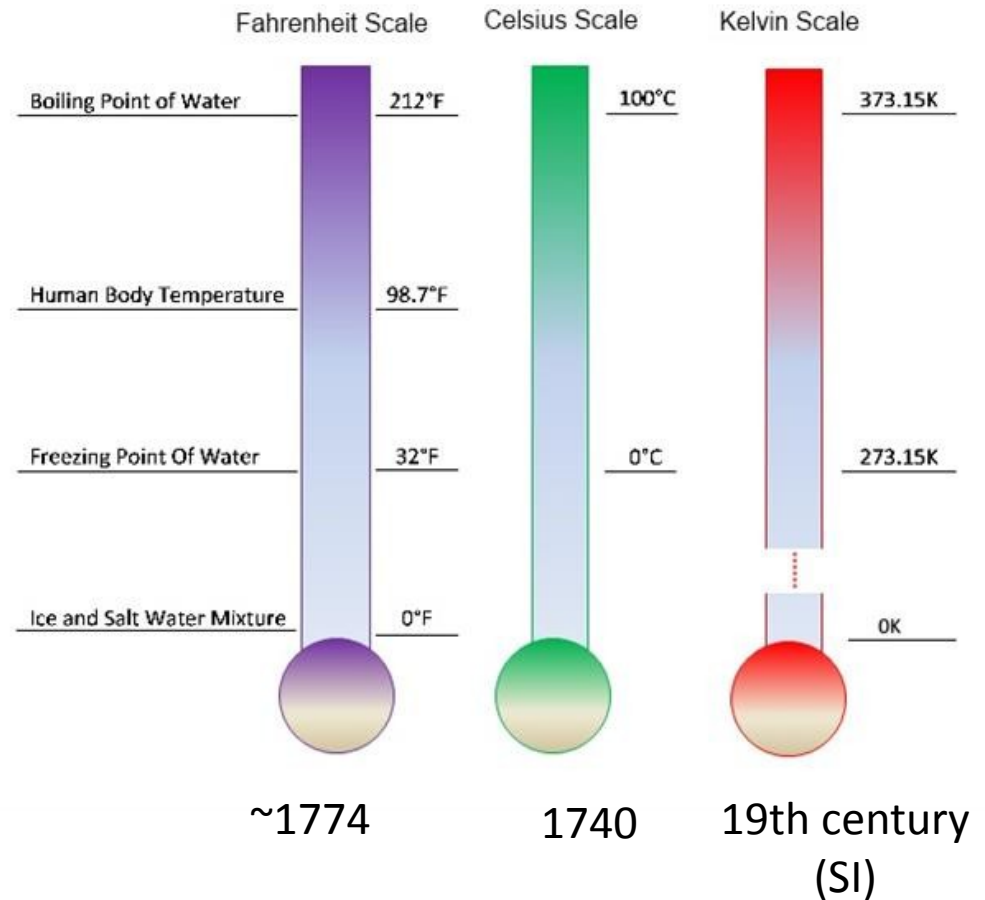
- **Definition:** physical quantity characterizing the average motion of molecules
 - Temperature is the measure of the internal heat energy of a substance

Units

- over 35 different temperature scales had been developed by the 18th century
- Kelvin scale:** Absolute zero is the point at which a system no longer contains any thermal energy and all molecular movement ceases

$$T_{(^{\circ}\text{C})} = (T_{(^{\circ}\text{F})} - 32) \times 5/9$$

$$T_{(^{\circ}\text{C})} = T_{(^{\circ}\text{K})} - 273.15$$



Measurements

- thermal balance with the environment



radio sounding



meteorological
shelter



satellite

Which factors determine temperature at a point?

■ Radiative factors

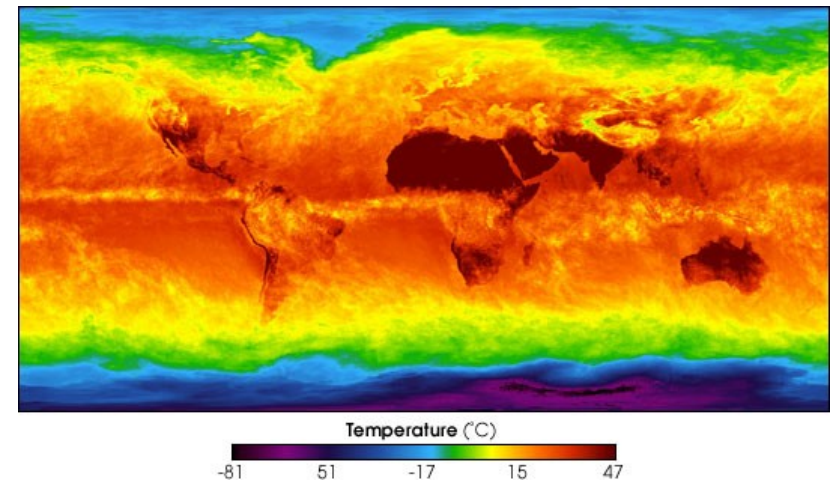
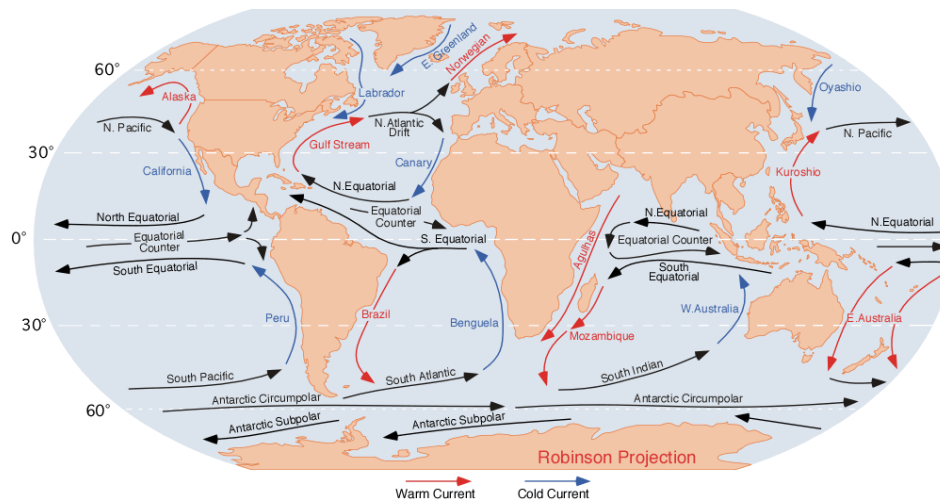
- Insolation (tilt of the earth, latitude, albedo, GHG, cloud)

■ Spatial and geographic factors

- where it is located horizontally and vertically

■ Dynamic & thermodynamic factors

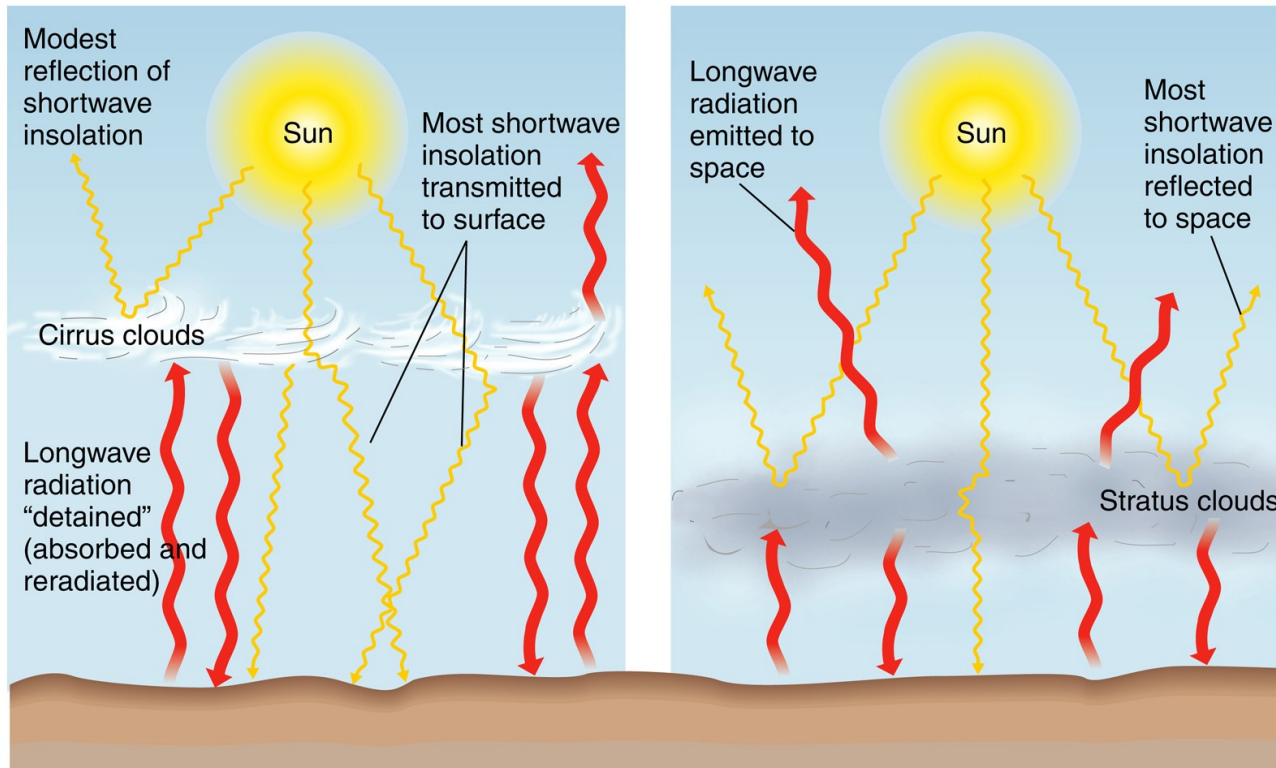
- Advective (wind) heat transfer, vertical motion
- Moisture content
- Latent heat phase changes
- Ocean currents
- Urban heat islands



Cloud coverage

1. **Q1:** Cloudy conditions predominate: smaller or larger temperature range?
2. **Q2:** Clear conditions predominate: smaller or larger temperature range?
3. **Q3:** Stratus type clouds cool surface temperatures better or weaker than cirrus clouds?

Cloud coverage



(a) High clouds: net greenhouse forcing and atmospheric warming

(b) Low clouds: net albedo forcing and atmospheric cooling

Thermodynamic Equation

$$\frac{\partial T}{\partial t} = -V \cdot \nabla T - w (\Gamma_d - \gamma) + \frac{1}{c_p} \Delta Q$$

Local
Temp.
change

Horizontal
adv.

Vertical
motion

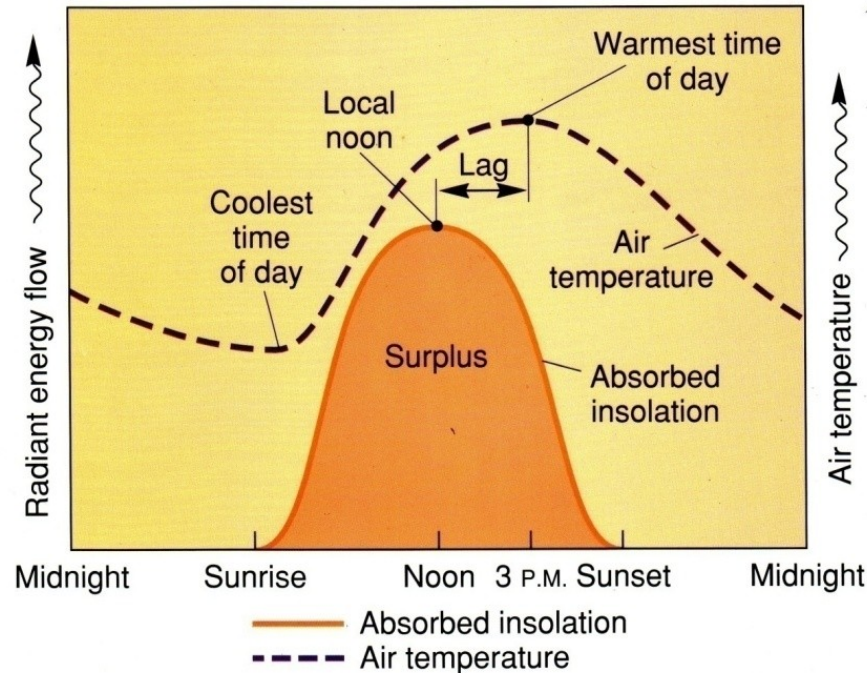
Diabatic
heating
(from radiation,
conduction,
latent heat)

Temporal controls on temperature

- **Q4:** At what time of day do the maximum daily temperatures generally occur?
- **Q5:** At what time of day do the minimum daily temperatures generally occur?
- **Q6:** Which month is the coldest/hottest month in the Northern Hemisphere?

Temporal controls on temperature

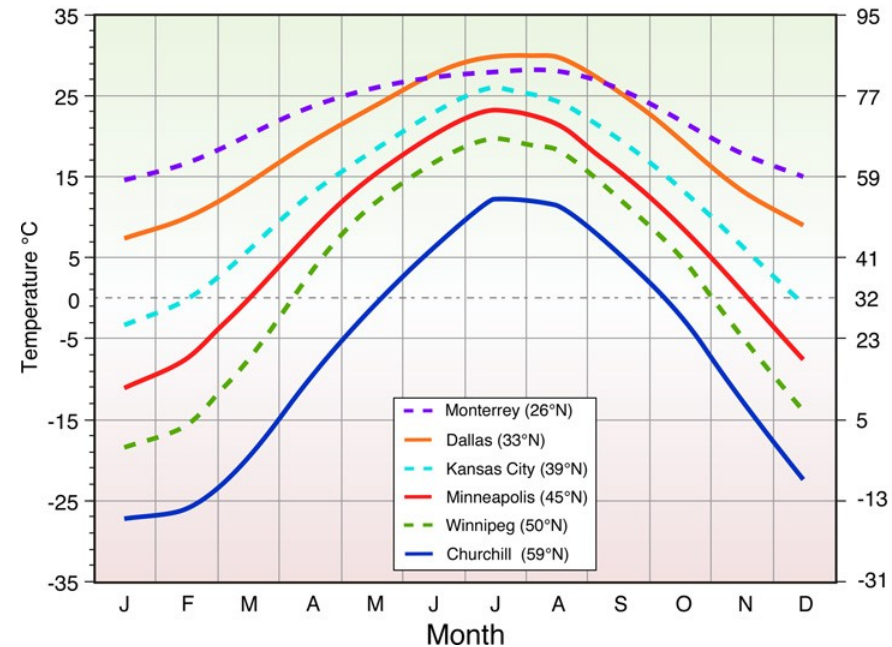
- Max temps occur not at solar noon but 2-4 hours after solar noon. **Why?**
- Minimum temperatures occur just after sunrise. **Why?**



Seasonal time lags

- **Dec 21st**: shortest day of the year. However, coldest temperatures are in Jan and Feb
 - ✧ it takes until Jan and Feb for all remaining LW radiation to be lost
- **June 21st**: longest day of the year. But warmest temperatures are in Aug
 - ✧ peaks in incoming SW radiation in June
 - ✧ LW re-emission reaches maximum in August

Latitude influence on mean monthly temperature. Mean monthly temperatures of 6 Northern Hemisphere locations with different latitudes (©Michael Pidwirny)



Adiabatic process and dry adiabatic lapse rate

Adiabatic process:

- An adiabatic process is a thermodynamic process in which **no heat or mass** is exchanged between a system (such as an air parcel) and its surrounding environment
- i.e. **$dQ=0$** where dQ represents the net heat added to or removed from the system

Adiabatic process and dry adiabatic lapse rate

Dry adiabatic lapse rate (Γ_d)

- the rate at which an unsaturated (dry) parcel of air cools as it rises, or warms as it sinks, solely due to adiabatic expansion or compression

Q. Prove that: $\Gamma_d = g/c_p$

The First Law of Thermodynamics & Hydrostatic equation

- $dq = c_p dT - \alpha dp$

where $\alpha = 1/\rho$ is the specific volume (volume per unit mass) and c_p is the specific heat at constant pressure

- & the hydrostatic equation $\rightarrow$

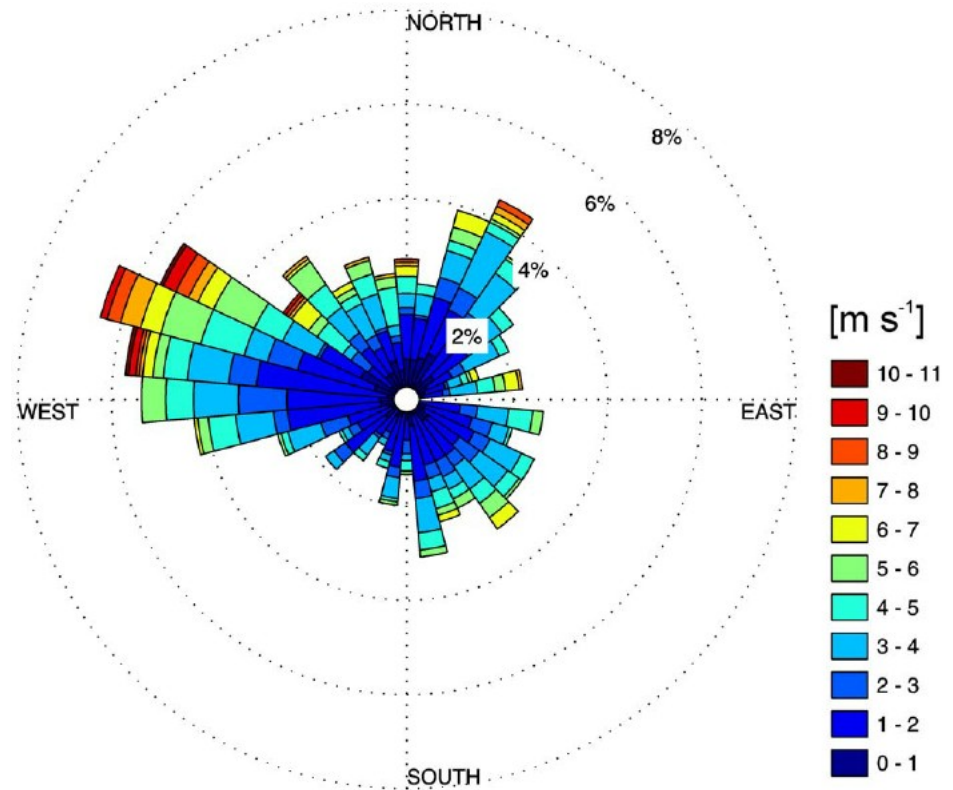
II. WIND

Direction

- Direction from where the wind comes with respect to the true North
- Expressed in sectors or degrees on **wind rose**



Wind vane



Wind rose for the Barrie Airport Weather Station for August and September 2011

doi:10.1371/journal.pone.0057444.g002

Speed (or intensity)

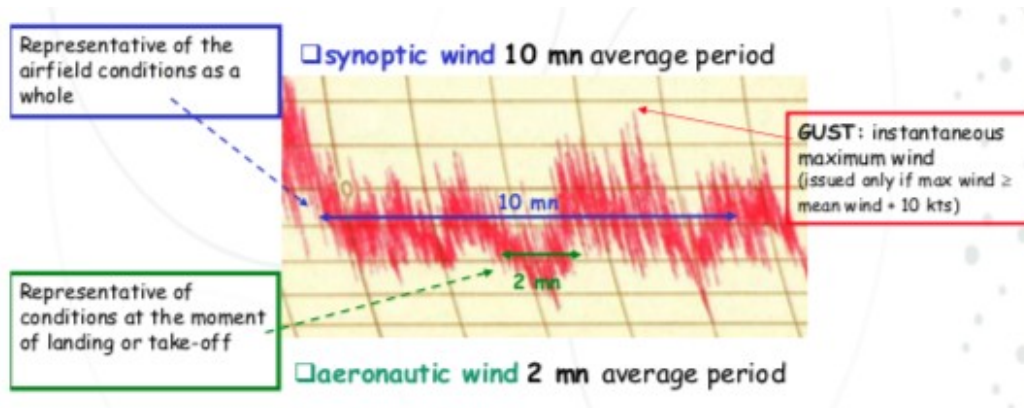


Surface continuous recording of instantaneous speed and direction

- Wind speed is expressed in m/s, kt (knot) or km/h
- E.g. 1 m/s $\sim$ 2kt $\sim$ 3,6 km/h

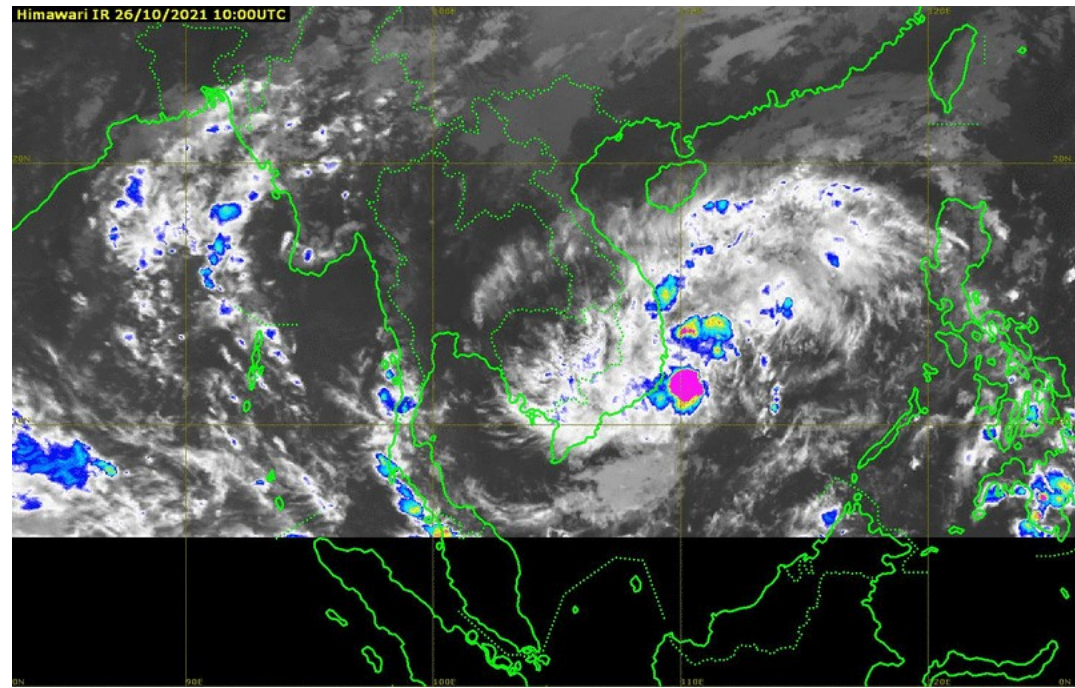
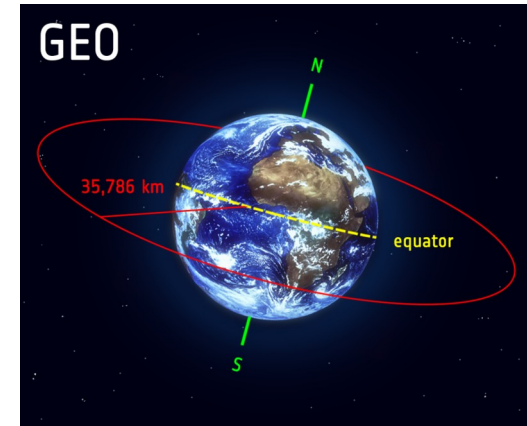
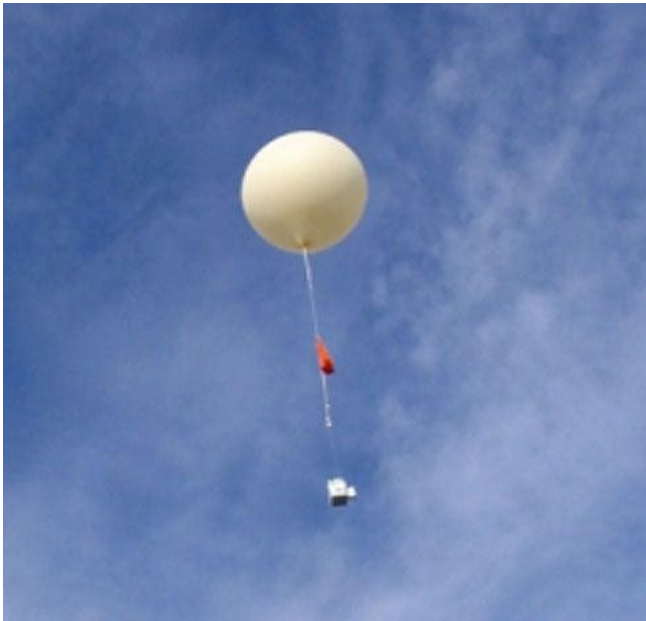
For tropical cyclones classification in NW Pacific (for 1-minute maximum sustained winds)

- Tropical depression: < 17 m/s
- Tropical storm: 18–32 m/s
- Typhoon: 119–239 km/h (33–66 m/s)
- Super typhoon: > 239 km/h



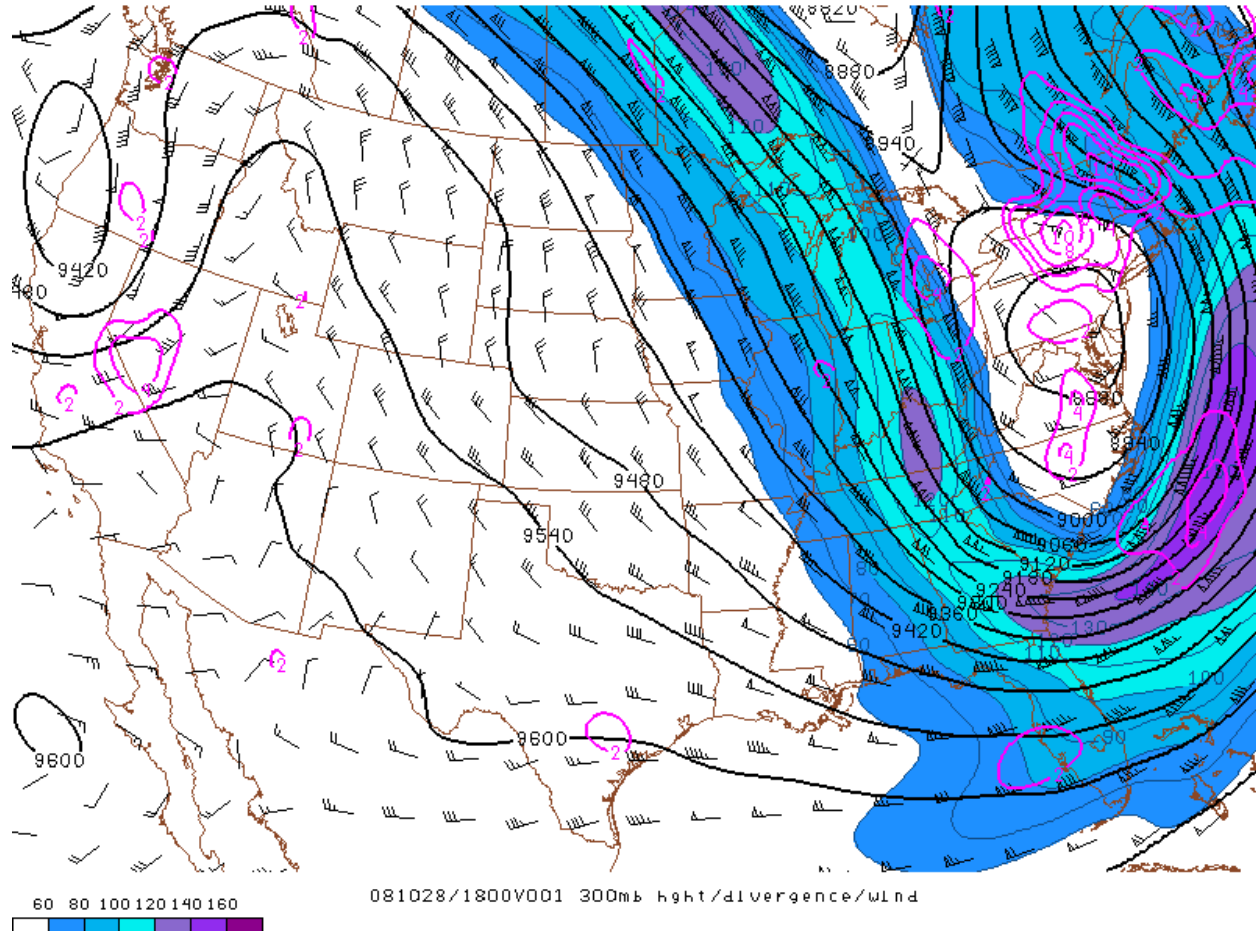
Wind in altitude

- Upper-air balloon (radiosonde) – tracked by satellite or by radar
- Cloud motion – tracked by satellite
- Measurement by aircraft



Wind mapping

- Isotachs: line of equal value of the wind speed



http://www.erh.noaa.gov/btv/events/28Oct2008/300mb_isotachs_28_18z.gif

Wind rose

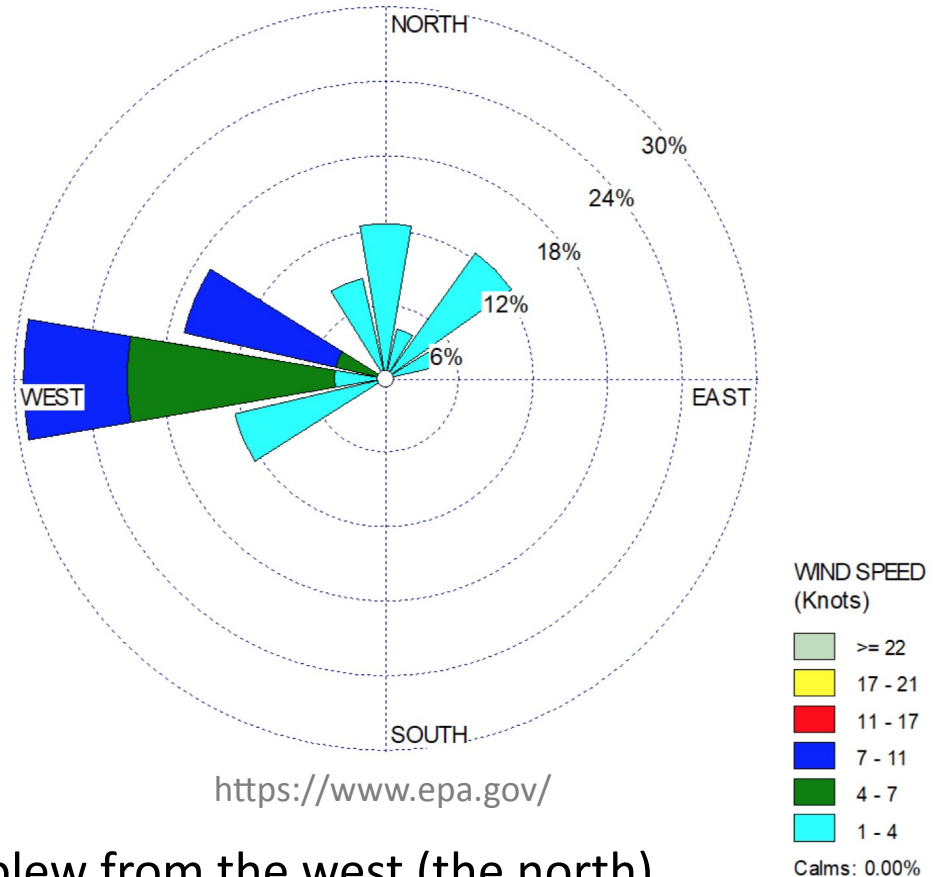
- a succinct view of how wind speed and direction are typically distributed at a particular location

How to read

- The circular format --> the direction the winds blew from
- The length of each "spoke" around the circle --> how often the wind blew from that direction

Example

- during the sampling period the wind blew from the west (the north) 30% (12%) of the time
- the longest spoke --> the wind blew from the west at speeds:
 - between 1-4 knots (light blue) about 4% of the time,
 - 4-7 knots (dark green) about 18% of the time
 - 7-11 knots (dark blue) about 7% of the time



Wind rose

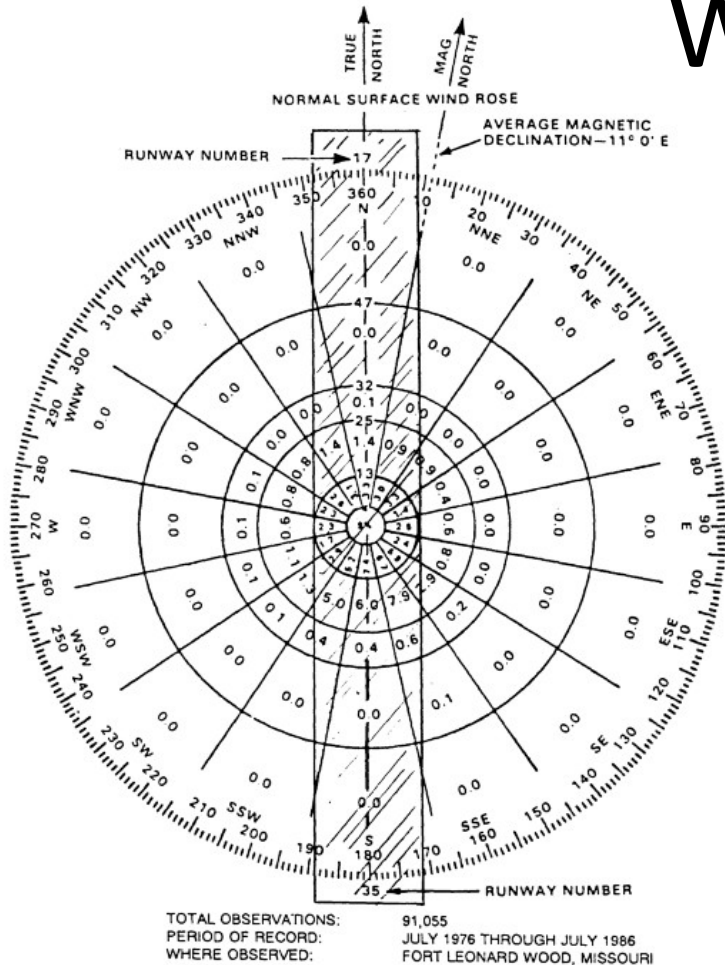


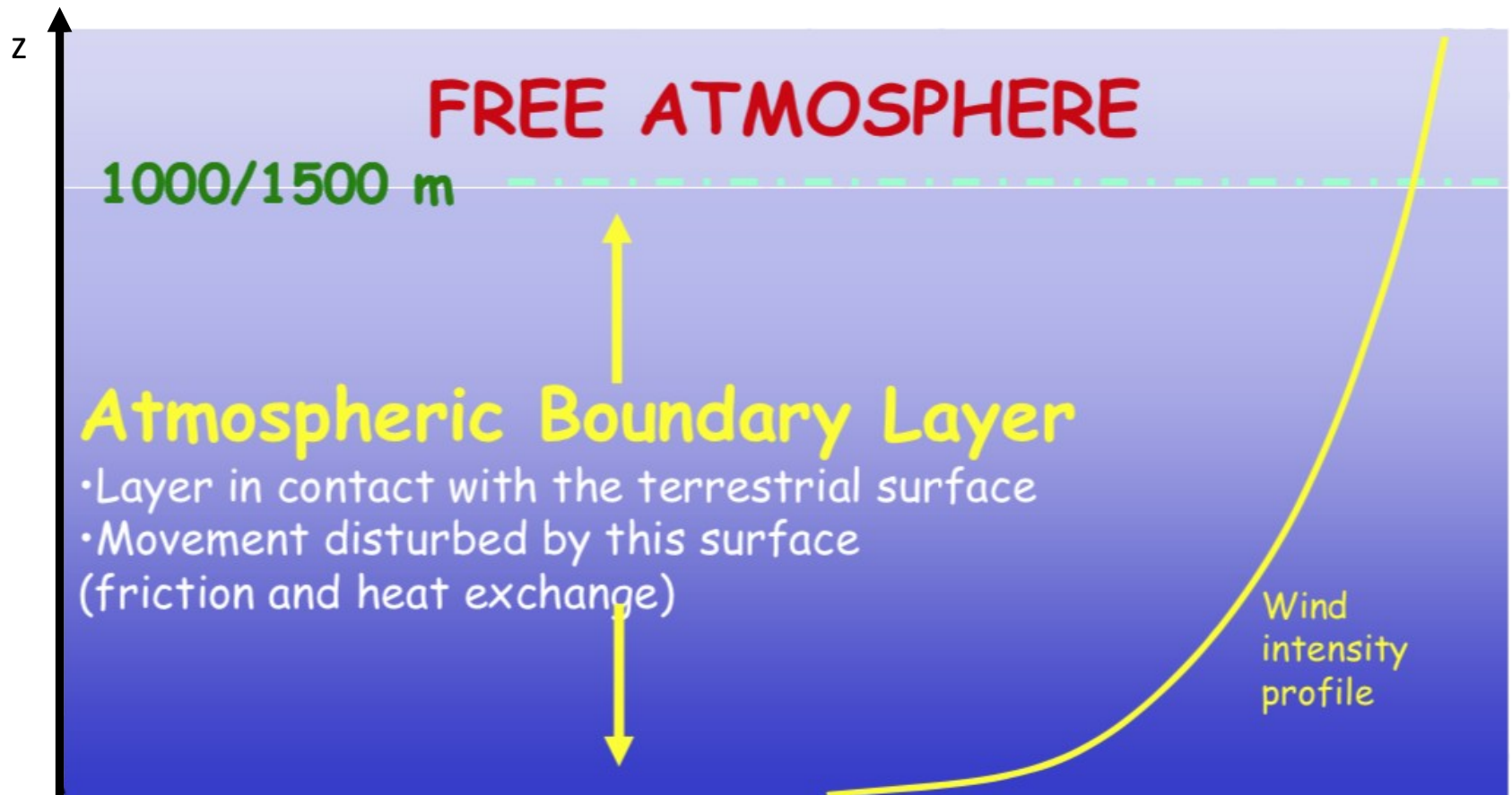
Figure 11-10. Determination of runway alignment by wind-rose analysis

Application: find the orientation of the runway to achieve the desired wind coverage.

- Use long-term data wind rose (e.g. 10 years)
- Runways are always orientated in the direction of prevailing winds:
 - **Take-off:** wind helps in generating the lift
 - **Landing:** wind helps in generating the drag

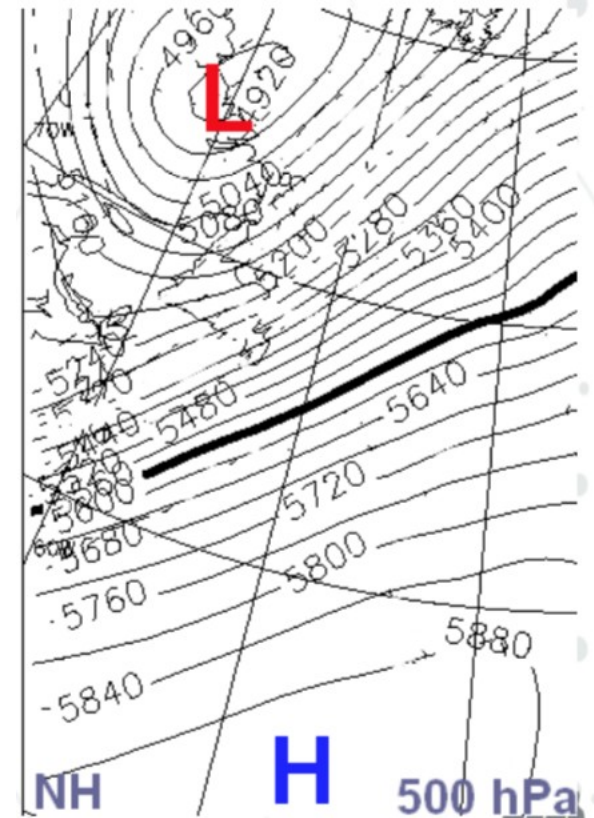
<https://www.globalsecurity.org/>

Wind in free atmosphere and Atmospheric boundary layer (ABL)

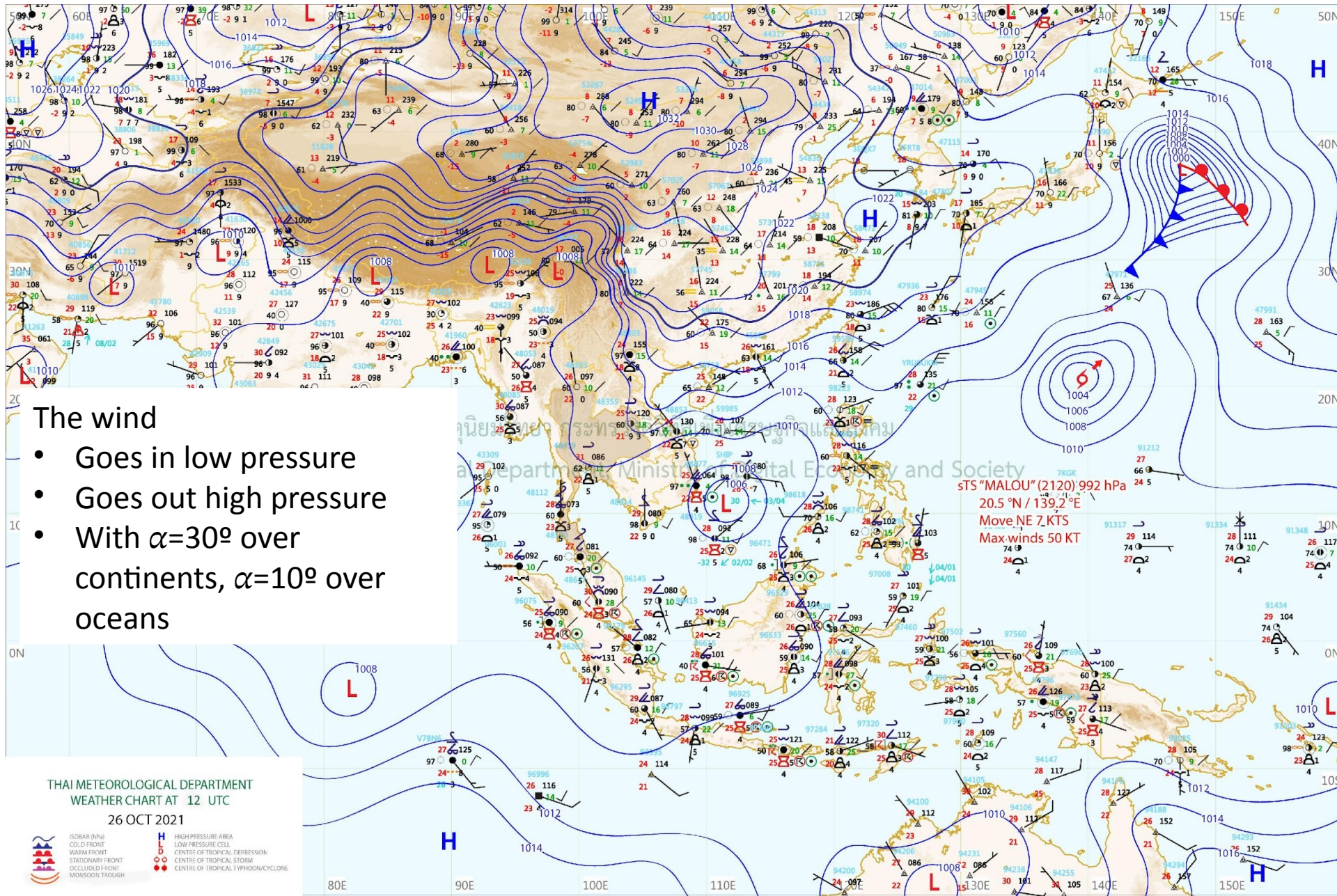


Buys-Ballot rules

- **In free atmosphere**, the wind is parallel to isobars (or isohypses)
- **In the Northern Hemisphere**, you stand with **your back to the wind**, pressure (or altitudes) is **lower on your left hand** than your right (Buys-Ballot, 1857)
- Wind's intensity is **inversely proportional** to the isohypses (or isobars) spacing



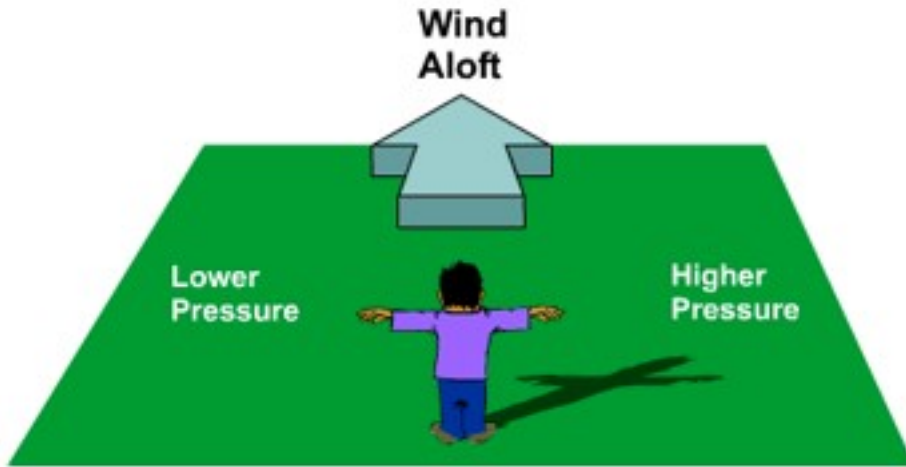
Close to the surface



Aloft:

Forces:

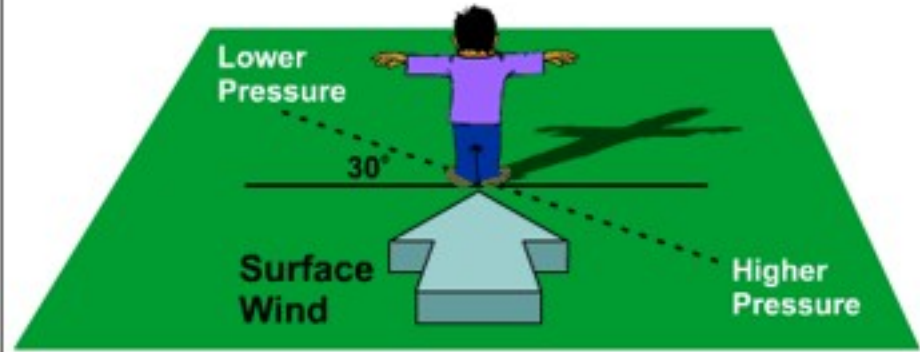
- pressure gradient,
- Coriolis,



Near the surface:

Forces:

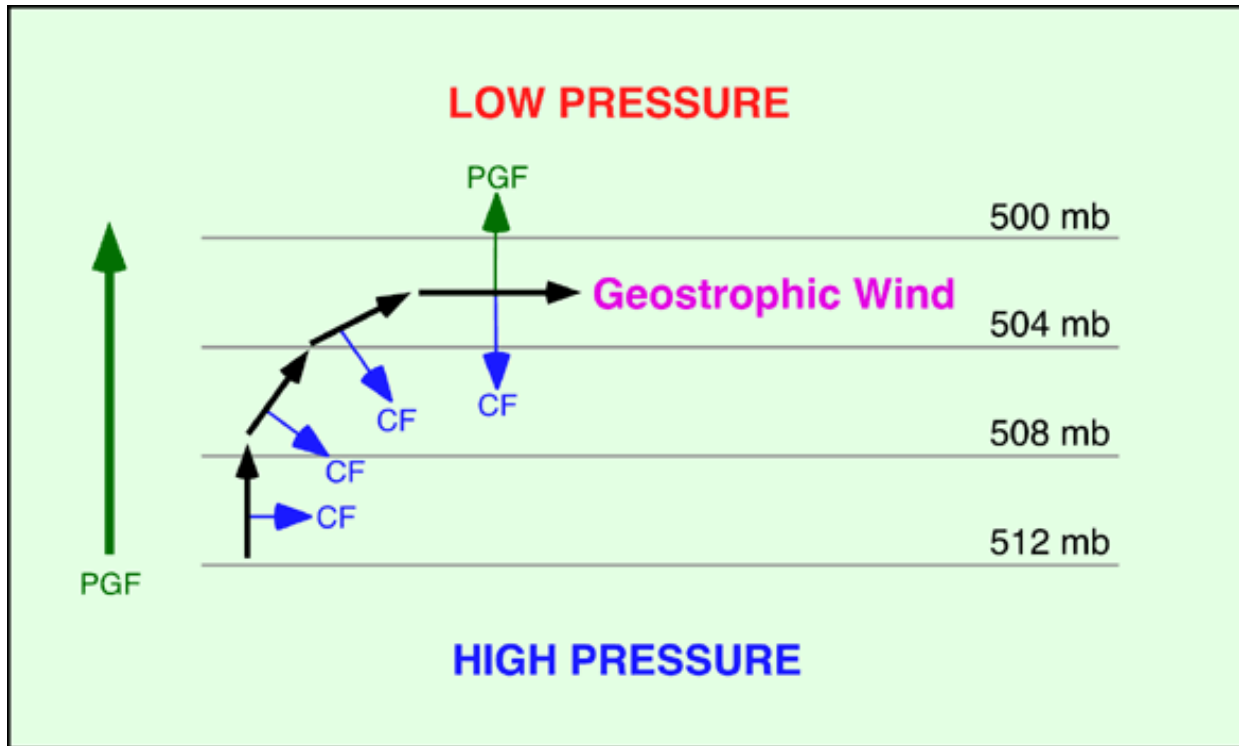
- pressure gradient,
- Coriolis,
- & frictional forces



<https://www.e-education.psu.edu/meteo3/node/2226>

- In the NH, you stand with your back to the wind, pressure is lower on your left hand than your right (Buys-Ballot, 1857)

Geostrophic wind (aloft)

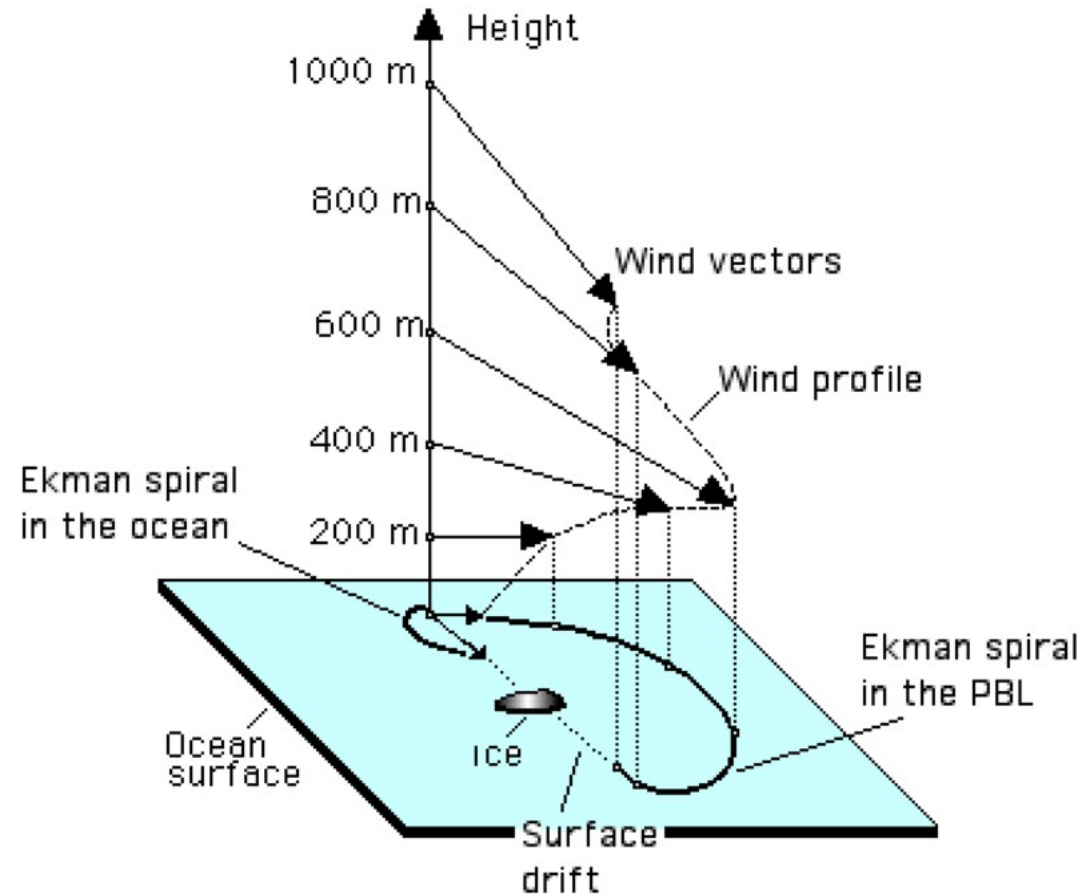
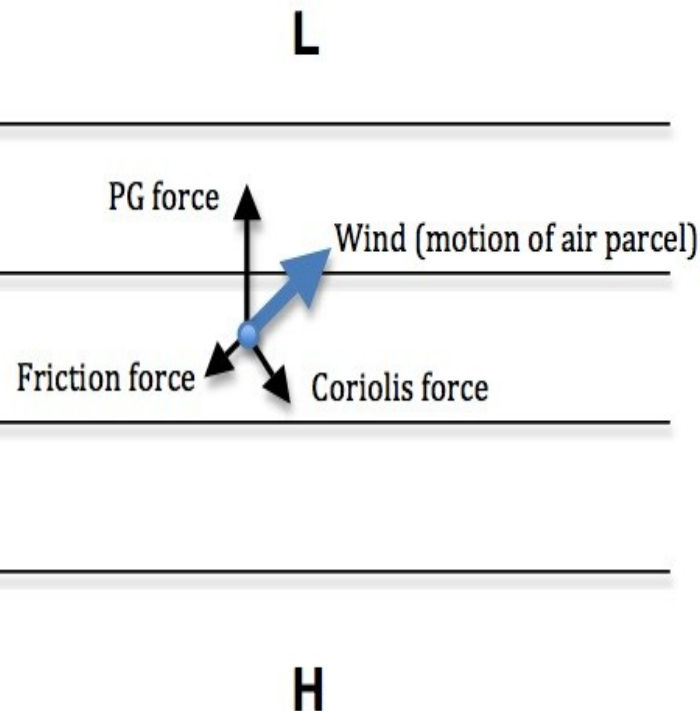
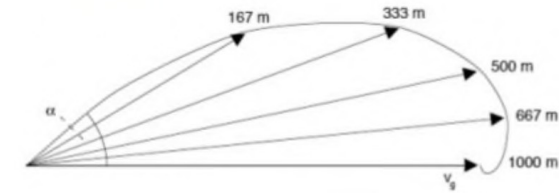


With $f = 2\Omega \sin \varphi$
the Coriolis parameter

<https://richhoffmanclass.com/chapter6.html>

Winds in the Atmospheric Boundary Layer

- Due to friction --> going down, the wind slows down and turn toward the low pressure system

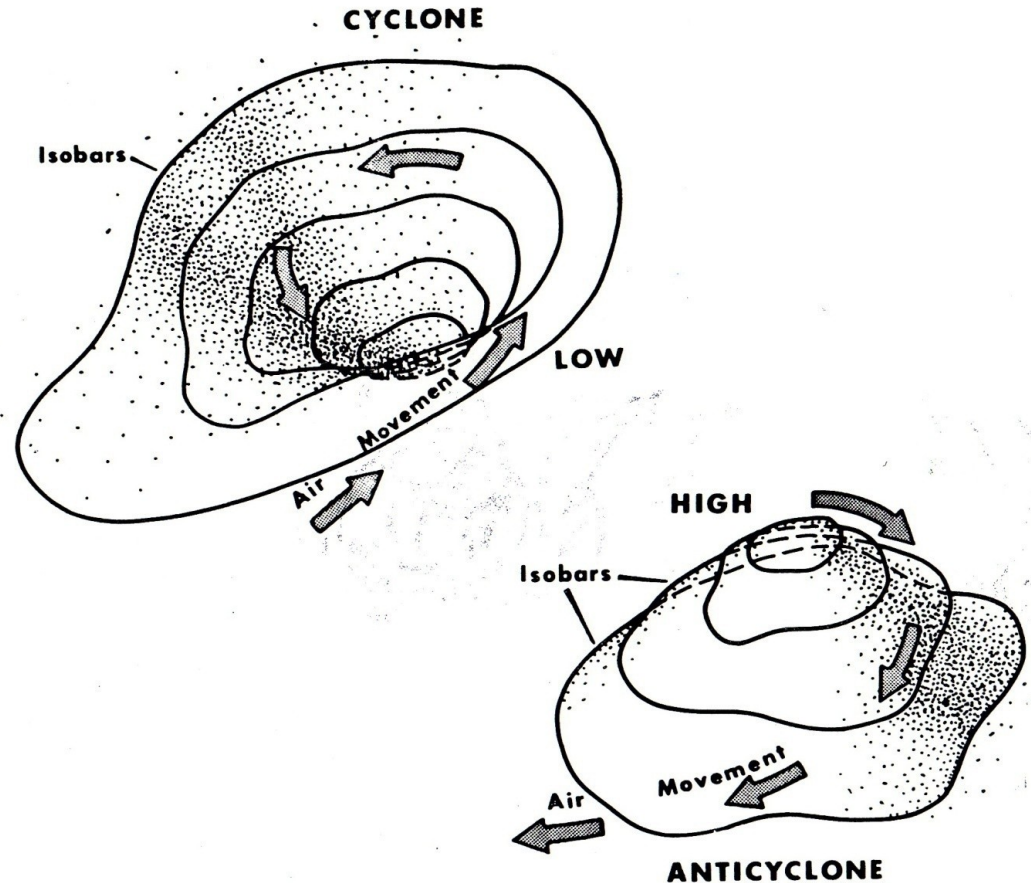


- Over ocean surface: $V_{10} \sim 70\% V_{1000}$
- Over continental surface: $V_{10} \sim 50\% V_{1000}$

Ekman spiral

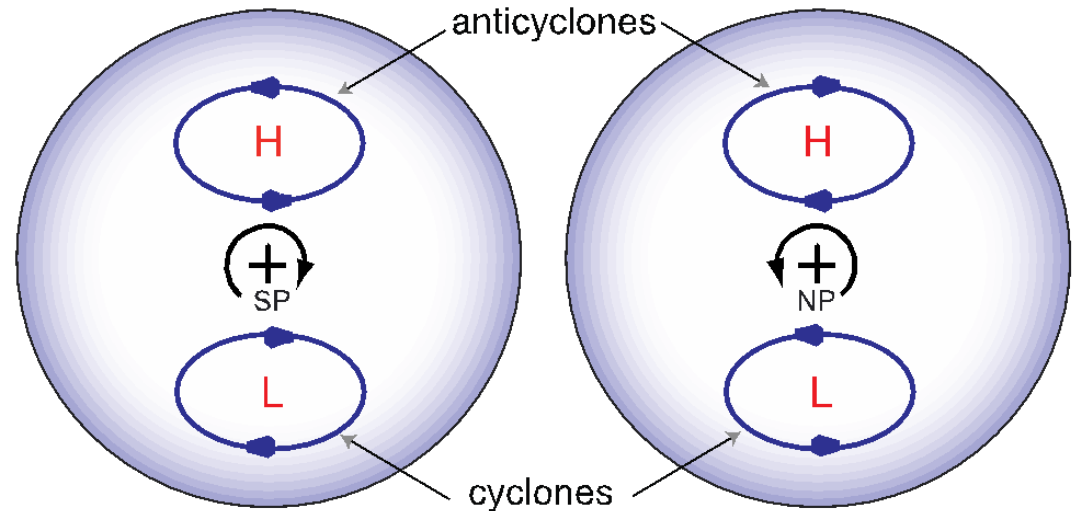
III. Pressure systems

- Two types: high and low
- **Low**: associated with clouds and instability.
- **High**: associated with clear conditions and stability



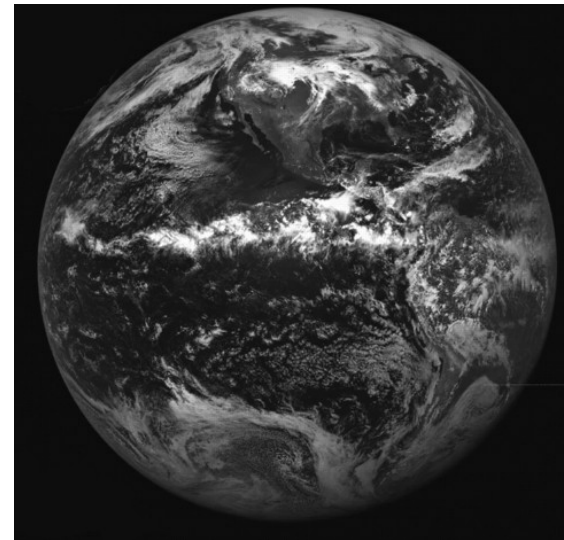
Pressure systems

- High-pressure centers or highs (H)
- Lows (L)
- Except for the equatorial belt (10S-10N), the winds observed in the atm. closely parallel the isobars
- In the NH, you stand with your back to the wind, pressure is lower on your left hand than your right (Buys-Ballot, 1857)
- **Equatorial belt:** directly across the isobars from higher toward lower pressure.



Blue arrows indicate the sense of the circulation around highs (H) and lows (L) in the pressure field, looking down on the South Pole (left) and the North Pole (right). Small arrows encircling the poles indicate the sense of the Earth's rotation.

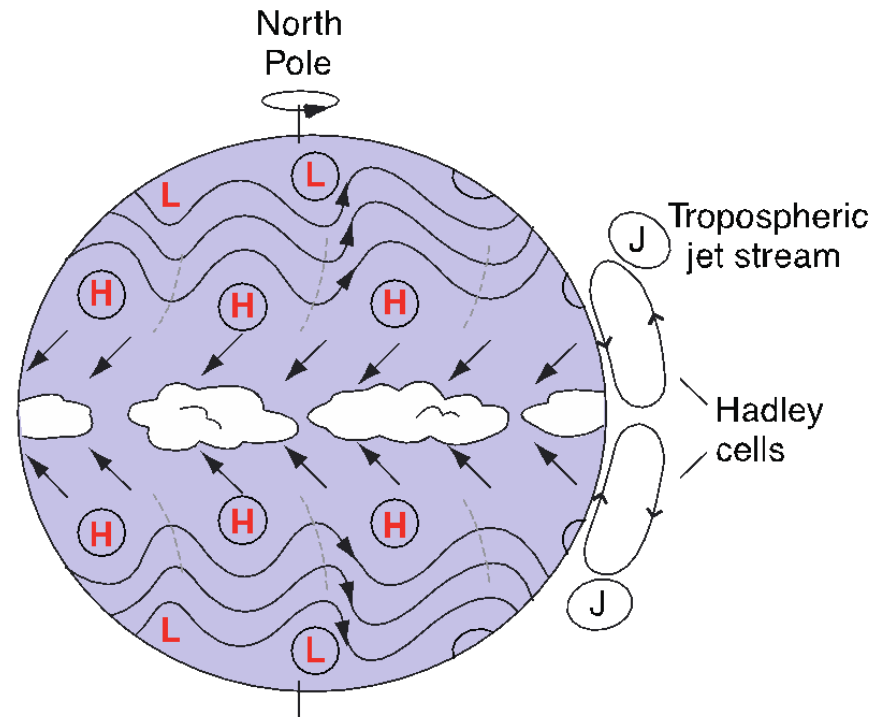
Which pressure systems



Aqua-planet

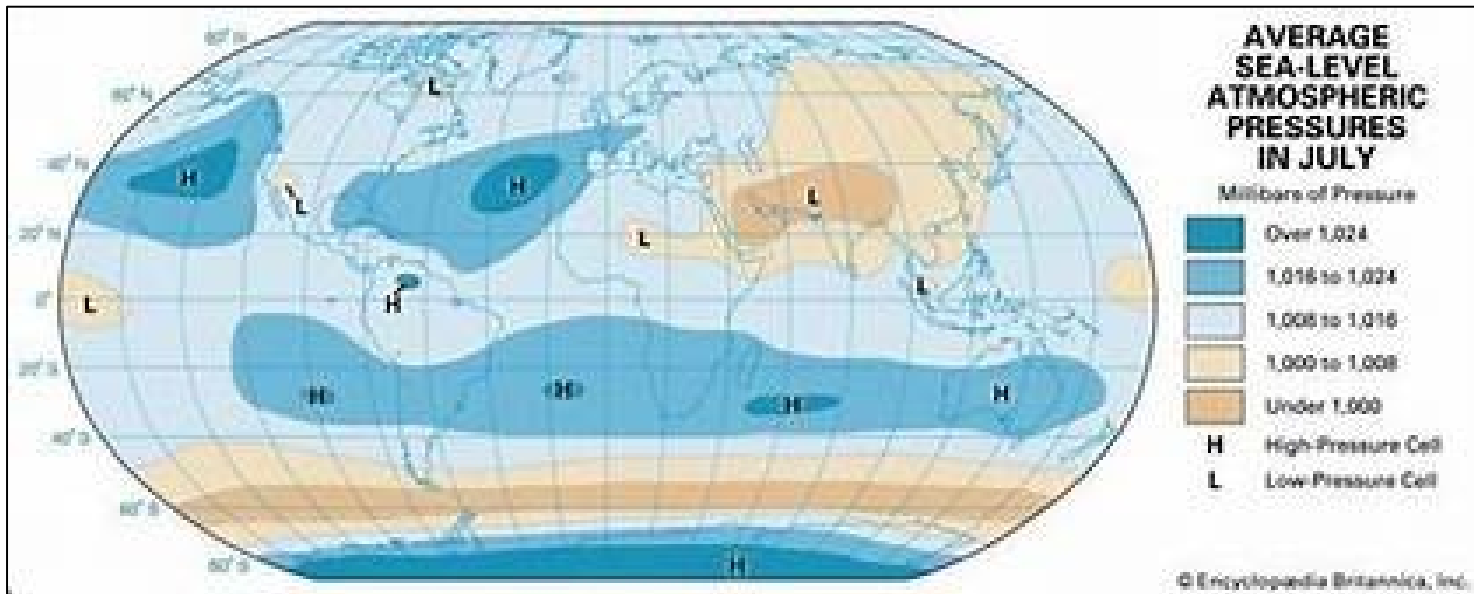
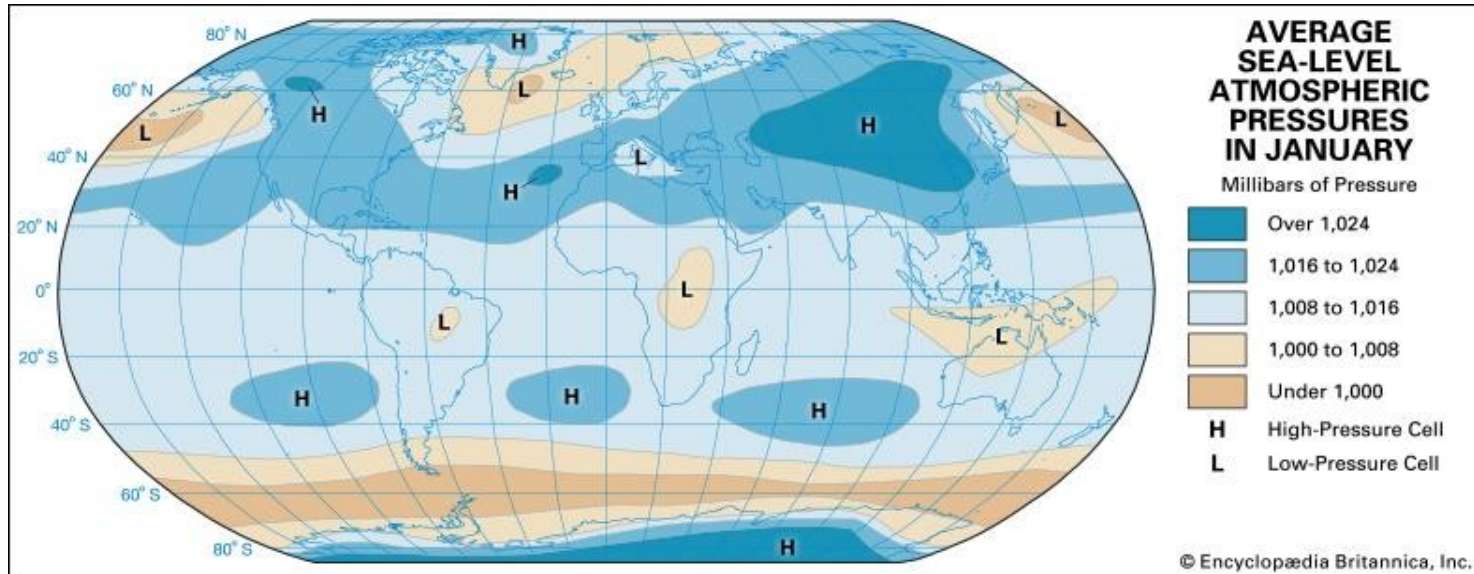
The observed surface wind field

- ✧ Tropical circulation: trade wind
Hadley cells, characterized by:
 1. **Equator-ward** flow in the boundary layer
 2. **Rising motion** in the equatorial zone
 3. **Poleward** return flow in the tropical upper troposphere
 4. **Sinking motion** in the subtropics

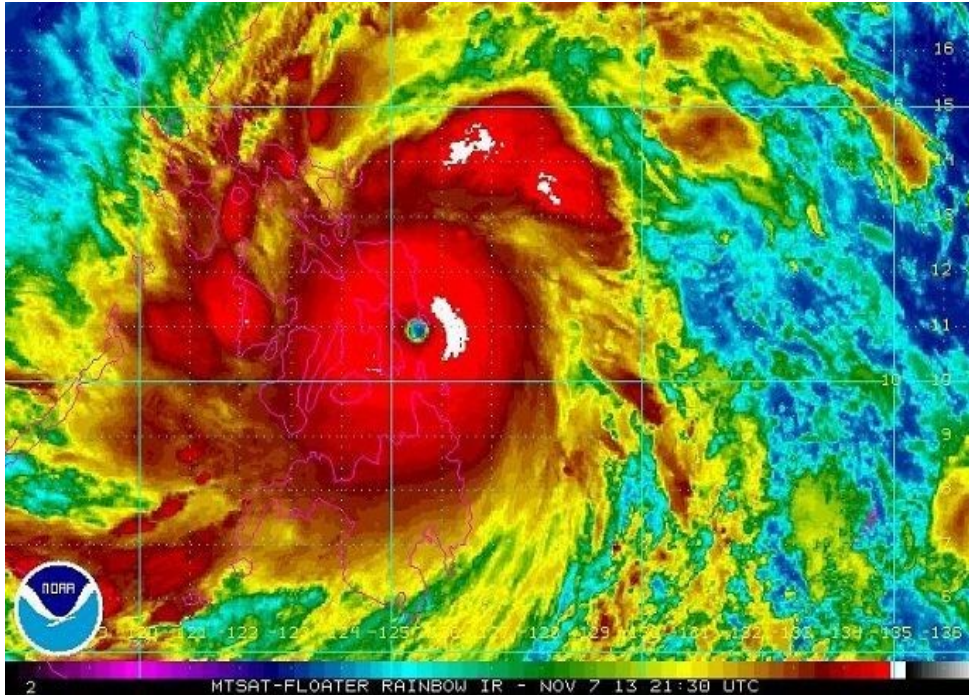


Schematic depiction of sea-level pressure isobars and surface winds on an idealized *aqua planet*, with the sun directly overhead on the equator. The rows of H's denote the subtropical high-pressure belts, and the rows of L's denote the subpolar low-pressure belt. Hadley cells and tropospheric jet streams (J) are also indicated.

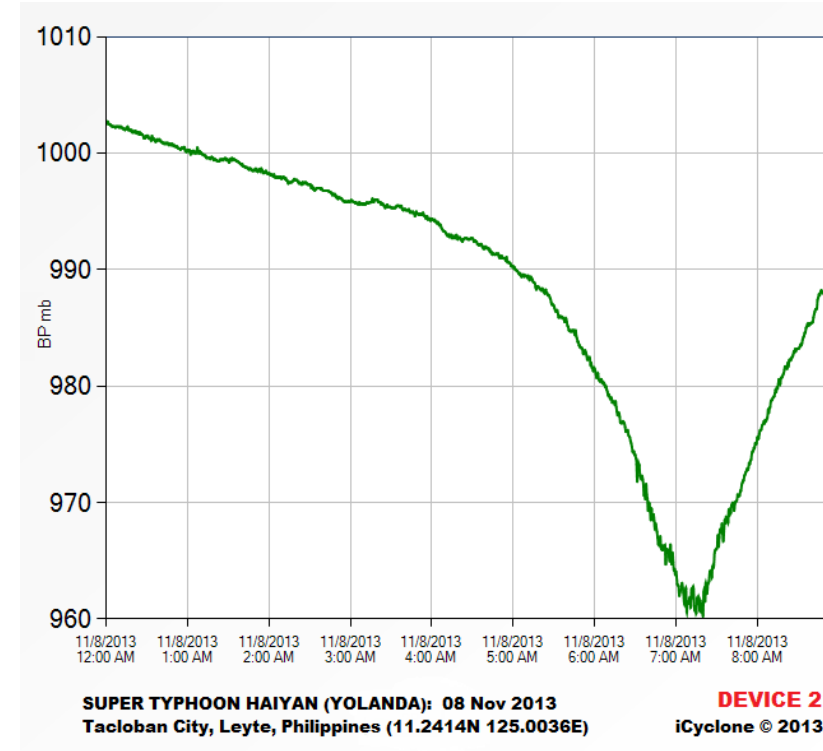
Real world



Extreme pressure values



Super Typhoon Haiyan as it made landfall in the Philippines, 7/11/2013



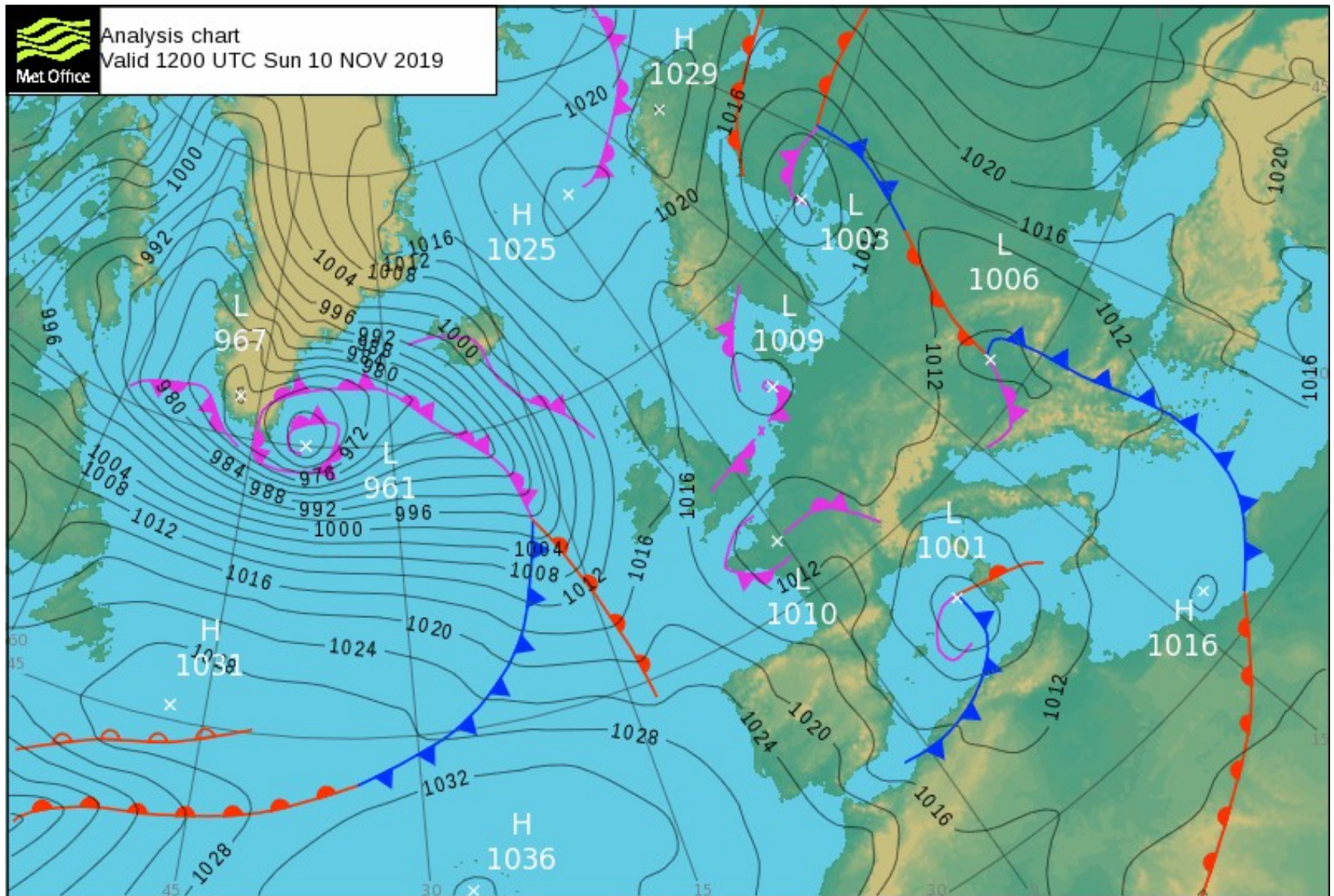
Representation of the pressure field

- at sea level: plot of **isobars**
- at upper-air: plot of **isohypses** (contour lines)
 - an isobar connects points of equal pressure
 - an isohypse connects points of equal **geopotential height** (vertical coordinate referenced to Earth's mean sea level)

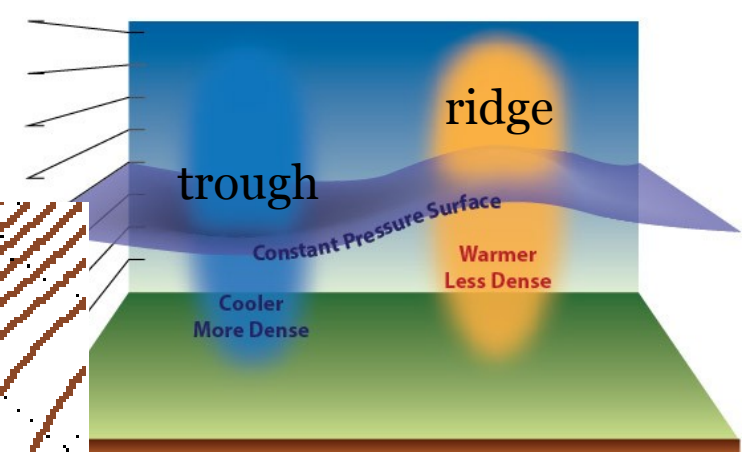
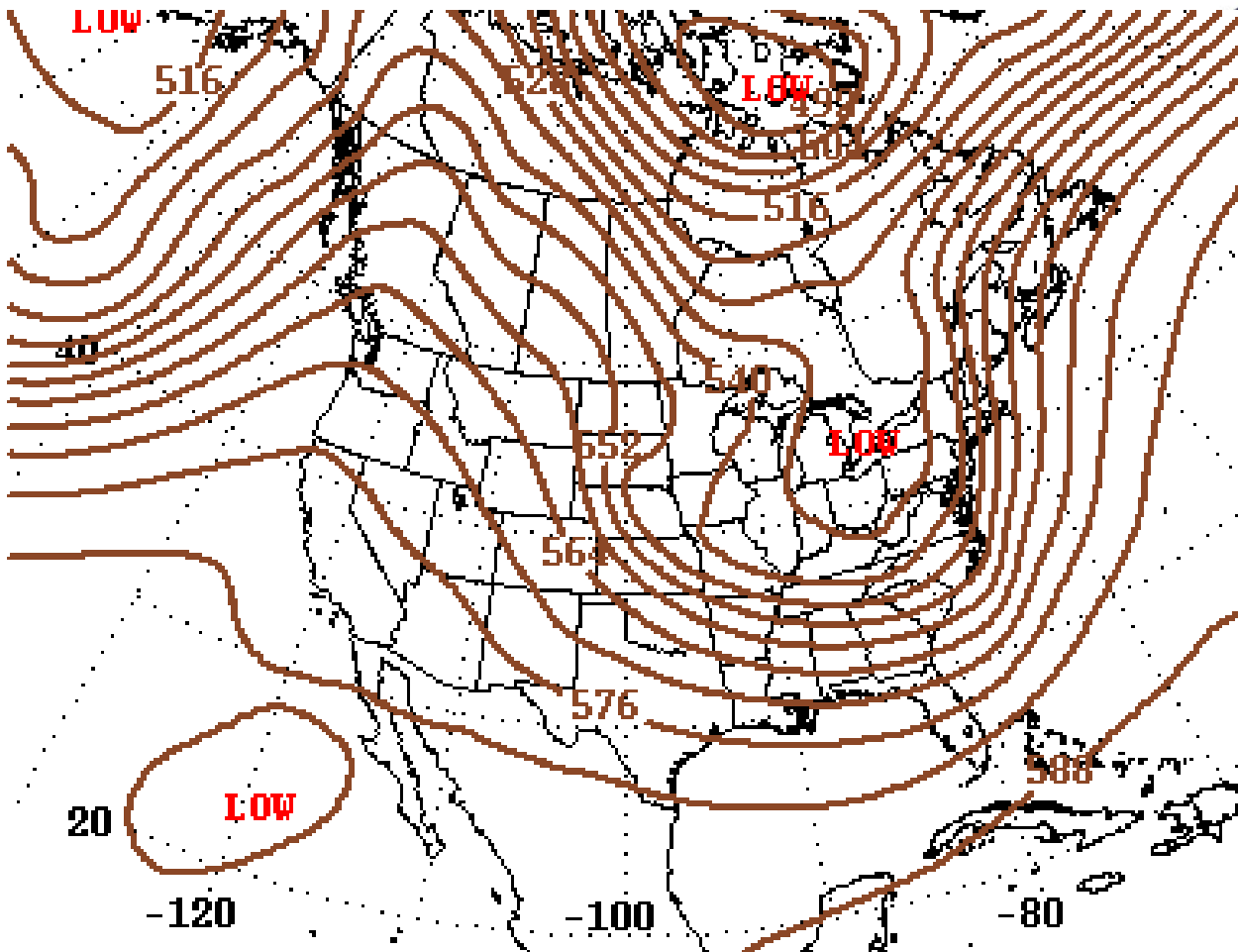
Geopotential height approximates the actual height of a pressure surface above mean sea-level. Therefore, a geopotential height observation represents the height of the pressure surface on which the observation was taken

Q: Calculate the geopotential height of the 850-hPa pressure surface when the pressure at sea level is 1014 hPa. The scale height of the atmosphere may be taken as 8 km.

At sea level: plot of isobars



Upper air analysis chart



A sample 500 millibar upper air chart. The gradient is the largest where the brown lines are closest together.

Practice #8: Thermal lag

- Model incoming solar radiation (SW_{in}) as a sine wave that peaks at solar noon ($t=12$) at 800 W/m^2 and is zero at night.
- Model outgoing terrestrial radiation (LW_{out}) using the Stefan-Boltzmann law (σT_g^4)

The system having a ground (temperature T_g) and a 2-m near surface air layer (temperature T_a), with the thermal inertia of ground and air being $2.0e5$ and $1.0e4$ respectively.

The thermodynamic equations are:

$$\frac{dT_{ground}}{dt} = \frac{SW_{in} - LW_{out} - Q_H}{C_{ground}}$$

$$\frac{dT_{air}}{dt} = \frac{Q_H}{C_{air}}$$

Q_H is the sensible heat flux,
with $h_{transfer}=15 \text{ Wm}^{-2}\text{K}^{-1}$.

$$Q_H = h_{transfer} \cdot (T_{ground} - T_{air})$$

- Time step $dt=60s$
- At $t=0$ (midnight), initial temperatures for both ground & air equal 18°C

Q. Plot the radiation curves alongside the temperature curve to show the exact time of the maximum temperature.